the value chain

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SAP Inside Track Belgium

Testing CAP apps locally made easy with the cds-launchpad-plugin

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Agenda – What can you expect

- How this project got started
- Cds-launchpad-plugin overview
- Incorporating the cds-launchpad-plugin in your own projects
- Demo
- Wrap-up
- Q&A

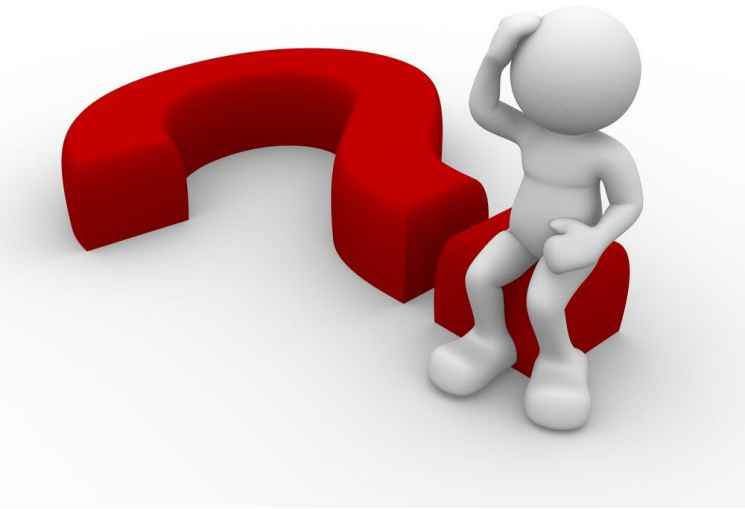
How this project got started

How this project got started – Personal interest & observations

- Working on multiple SAP Cloud Application Programming Model-based projects
- Performing same actions in each project
  - Copy Fiori Launchpad Sandbox HTML page
  - Setting up a FioriSandboxConfig.json file
- Learning TypeScript
- Looking into reusable component development with Nodejs

How this project got started – Some questions kept popping up

- Why am I repeating the same process for each and every project?
- What if for some reason the HTML for the Fiori Launchpad Sandbox needs to change?
- Why am I maintaining the config file, while this information is already available in the project?
- How can I easily validate my project's UI configuration?



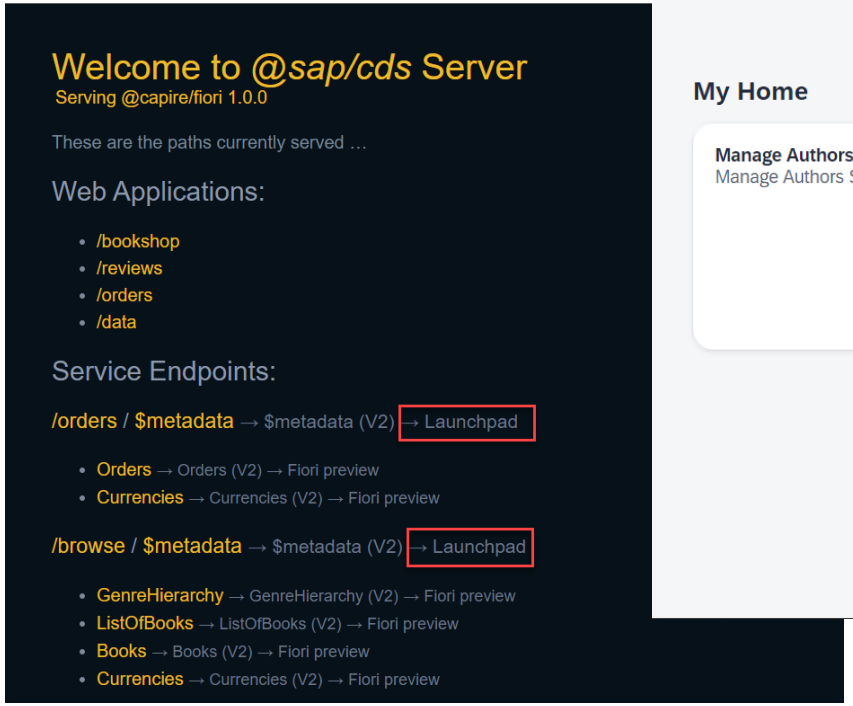
How this project got started – The idea

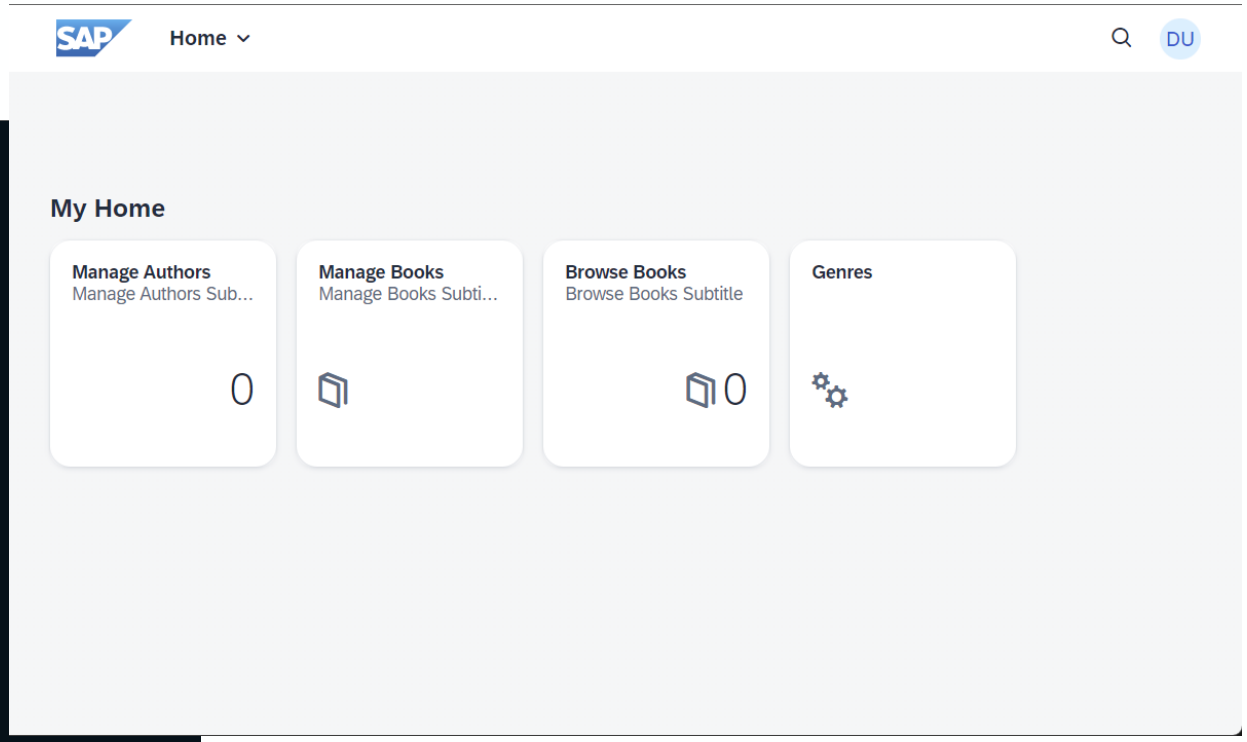
- Reusable solution
- Generate Fiori Launchpad Sandbox
- Leverage only SAP CAP / SAPUI5 standard project configuration
- Allow project specific configuration
  - SAPUI5 theme
  - SAPUI5 version
- Integrated in CAP development tools
- Built using TypeScript (personal challenge ☺)
- Open-source

Cds-launchpad-plugin overview

Cds-launchpad-plugin overview – What is it?

- Plugin for SAP Cloud Application Programming Model (@sap/cds)
- Compatible with Node.js-based CAP projects
- Generated Fiori Launchpad Sandbox for testing / demo purposes
- Plugin for CDS Server





Cds-launchpad-plugin overview – What it’s not?

- A plugin to use productively
  - No authorization management
  - No user management
  - No personalization
  - No theming
  - …
- A replacement for SAP Launchpad / SAP Work Zone Service

Cds-launchpad-plugin overview – Advantages

- Easy to install
- Improved developer efficiency
- Easy to integrate in your project (only a few lines of code needed)
- Implicitly validates your whole project configuration (in terms of UI)
- Easily test your SAPUI5 apps & plugins
- No manual configuration needed (if your project configuration is properly in place)
- Create your own HTML Launchpad Sandbox but leverage generated appconfig file
- Create your own appconfig file and merge it with the generated one

Incorporating in your own projects

Incorporating in your own projects – Overview

- Install using a single command
  - npm install --save-dev cds-launchpad-plugin
- Four important files for successfully using the plugin
  - server.js (CAP)
  - project.json (CAP)
  - manifest.json (SAPUI5)
  - i18n.properties (SAPUI5)

Incorporating in your own projects – server.js

- Activate the plugin in non-production environments
- Configuration object can be ommited

fiori > JS server.js > ...

```javascript
const cds = require("@sap/cds")

// install OData v2 adapter
const proxy = require('@sap/cds-odata-v2-adapter-proxy')
const proxyOpts = global.it ? { target:'auto' } : {} // for tests, set 'auto' to detect port dynamically
cds.on('bootstrap', app => app.use(proxy(proxyOpts)))

// activate launchpad plugin
if (process.env.NODE_ENV !== 'production') {
    const {cds_launchpad_plugin} = require('cds-launchpad-plugin');

    // Enable launchpad plugin
    cds.once('bootstrap',(app)=>{
        const handler = new cds_launchpad_plugin();
        app.use(handler.setup({theme:'sap_horizon', version: '1.105.1'}));
    });
}

module.exports = require('@capire/bookstore/server.js')
```

Incorporating in your own projects – project.json

- Ensure sapux section contains paths to SAPUI5 apps
- Should be automatically generated when using Fiori Tools

{} package.json ×

fiori > {} package.json > [ ] sapux

```
1  {
2    "name": "@capire/fiori",
3    "version": "1.0.0",
4    "dependencies": {
5      "@capire/bookstore": "*",
6      "@sap/cds": ">=5",
7      "@sap/cds-odata-v2-adapter-proxy": "^1.9.0",
8      "cds-launchpad-plugin": "^1.0.7",
9      "express": "^4.17.1",
10     "passport": ">=0.4.1"
11   },
       ▷ Debug
12   "scripts": {
13     "start": "cds run --in-memory?",
14     "watch": "cds watch"
15   },
16   "cds": {
17     "requires": {
18       "ReviewsService": {
19         "kind": "odata",
20         "model": "@capire/reviews"
21       },
22       "OrdersService": {
23         "kind": "odata",
24         "model": "@capire/orders"
25       },
26       "messaging": {
27         "[production]": {
28           "kind": "enterprise-messaging"
29         },
30         "[development]": {
31           "kind": "file-based-messaging"
32         },
33         "[hybrid!]": {
34           "kind": "enterprise-messaging-shared"
35         }
36       },
37       "db": {
38         "kind": "sql"
39       },
40       "db-ext": {
41         "[development]": {
42           "model": "db/sqlite"
43         },
44         "[production]": {
45           "model": "db/hana"
46         }
47       }
48     },
49     "hana": {
50       "deploy-format": "hdbtable"
51     }
52   },
53   "sapux": [
54     "app/admin-authors",
55     "app/admin-books",
56     "app/browse",
57     "app/genres",
58     "app/bookshopplugin"
59   ]
60 }
61
```

Incorporating in your own projects – manifest.json - Apps

- Ensure an inbound tile configuration is present
  - Semantic Object
  - Action
  - Title
  - Info (optional)
  - Subtitle (optional)
  - Icon (optional)

orders > app > orders > webapp > {} manifest.json > {} sap.ui5 > {} dependencies

```
1  {
2      "_version": "1.8.0",
3      "sap.app": {
4          "id": "orders",
5          "type": "application",
6          "title": "Order Books",
7          "description": "Sample Application",
8          "i18n": "i18n/i18n.properties",
9          "dataSources": {
10             "OrdersService": {
11                 "uri": "/orders/",
12                 "type": "OData",
13                 "settings": {
14                     "odataVersion": "4.0"
15                 }
16             }
17         },
18         "-sourceTemplate": {
19             "id": "ui5template.basicSAPUI5ApplicationProject",
20             "-id": "ui5template.smartTemplate",
21             "-version": "1.40.12"
22         },
23         "crossNavigation": {
24             "inbounds": {
25                 "intent1": {
26                         "signature": {
27                         "parameters": {},
28                         "additionalParameters": "allowed"
29                     },
30                     "semanticObject": "Orders",
31                     "action": "manage",
32                     "title": "{{appTitle}}",
33                     "info": "{{appInfo}}",
34                     "subTitle": "{{appSubTitle}}",
35                     "icon": "sap-icon://sales-order"
36                 }
37             }
38         }
39     },
40     "sap.ui5": {
```

Incorporating in your own projects – manifest.json - Plugins

- Ensure an inbound tile configuration is present
  - Semantic Object = Shell
  - Action = plugin
- Ensure Launchpad Configuration is present
  - Type = plugin

manifest.json 2, M

fiori > app > snowflakes > webapp > manifest.json > sap.app > crossNavigation > inbounds > intent1

```
1  {
2      "_version": "1.0.0",
3      "sap.app": {
4          "id": "be.fiddle.snowflakes",
5          "type": "component",
6          "title": "{{apptitle}}",
7          "description": "{{appdesc}}",
8          "i18n": "i18n/i18n.properties",
9          "applicationVersion": {
10             "version": "1.0.0"
11         },
12         "crossNavigation": {
13             "inbounds": {
14               "intent1": {
15                 "signature": {
16                     "parameters": {},
17                     "additionalParameters": "allowed"
18                 },
19                 "semanticObject": "Shell",
20                 "action": "plugin"
21               }
22             }
23         }
24     },
25     "sap.flp": {
26         "type": "plugin",
27         "config": {}
28     },
```

Incorporating in your own projects – i18n.properties

- Supports multiple languages
- Only needed if used in tile configuration

orders > app > orders > webapp > i18n > i18n.properties

```
# This is the resource bundle of itelo
# __ldi.translation.uuid=c3431418-9caf-11e8-98d0-529269fb1459

# JCI app descriptor contains lower case TITLE
appTitle=Bookshop Sample

# JCI app descriptor contains lower case DESCRIPTION
appSubTitle=Bookshop Sample Subtitle

# JCI app descriptor contains lower case DESCRIPTION
appDescription=Bookshop Sample Description

# JCI app descriptor contains lower case DESCRIPTION
appInfo=Bookshop Sample Info
```

Demo

Demo – Demo repository

- Available at https://github.com/geert-janklaps/cds-launchpad-plugin-demo
- Branches
  - 0-Base-Sample -> un-adapted bookshop sample
  - 1-Launchpad-Plugin-Base -> activated cds-launchpad-plugin + project corrected
  - 2-Launchpad-Plugin-SAPUI5-Launchpad-Plugin -> added snowflake plugin
  - 3-Launchpad-Plugin-Merged-Config -> project config file merged with generated config file
- Run sample with following command
  - npm run fiori
- If login needed
  - User: alice
  - No password

Wrap-up

Wrap-up – Contributors

- Sergei Haller (https://github.com/the-docta)
  - Multiple intents support
  - External / Merged appconfig file functionality
  - Dynamic tile support
- Jibbril (https://github.com/Jibbril)
  - Launchpad Plugin support

IDEAS TO IMPROVE THE PLUGIN ARE ALWAYS WELCOME!

EITHER BY CONTRIBUTING OR BY CREATING AN ISSUE IN GITHUB! ☺

Wrap-up – Overview

- Quick installation
- Easy to use
- No more manual configuration needed
- Leverages configuration that should be present anyway
- Ensures project configuration is correct in terms of UI
- Saves time / improves developer efficiency
- Future proof

Wrap-up – Intresting links

- GitHub: https://github.com/geert-janklaps/cds-launchpad-plugin
- GitHub Demo Repository: https://github.com/geert-janklaps/cds-launchpad-plugin-demo
- Solution Overview Blog: https://blogs.sap.com/2022/03/14/a-fiori-launchpad-sandbox-for-all-your-cap-based-projects-overview/

Q&A